

Report IV
Solving (sparse) system of linear equations (SLE)
with successive overrelaxation method (SOR)

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1. Introduction

The aim of this project was to solve a sparse system of linear equations obtained from discretization of the one-dimensional Poisson equation:

$$\frac{d^2V(x)}{dx^2} = -\rho(x)$$

Using the finite difference method, the differential equation is transformed into an algebraic system:

$$A\vec{v} = \vec{\rho}$$

where A is a tridiagonal matrix. Due to its sparsity, the system can be efficiently solved using iterative methods.

In this work, the Successive Over-Relaxation method is applied. The convergence of the method strongly depends on the relaxation parameter ω .

2. Description of the Numerical Task

The computational domain is discretized into $n = 500$ points with spacing:

$$\Delta = 1$$

The second derivative is approximated using the central difference:

$$\frac{d^2V}{dx^2} \approx \frac{V_{i+1} - 2V_i + V_{i-1}}{\Delta^2}$$

which leads to a tridiagonal system with:

- diagonal elements: $-\frac{2}{\Delta^2}$
- off-diagonal elements: $\frac{1}{\Delta^2}$

The right-hand side vector is defined as:

$$\rho_i = 0.1 \cdot \left[\sin\left(\frac{2\pi \cdot i}{n-1}\right) \right]^{10}$$

The system is solved using the SOR iterative algorithm:

$$v_i \leftarrow v_i + \frac{\omega}{d_i} (\rho_i - (Av)_i)$$

Calculations were performed for:

$$\omega = 1, 1.5, 1.9, 1.99$$

with stopping condition:

$$\|r\| < \varepsilon$$

3. Results

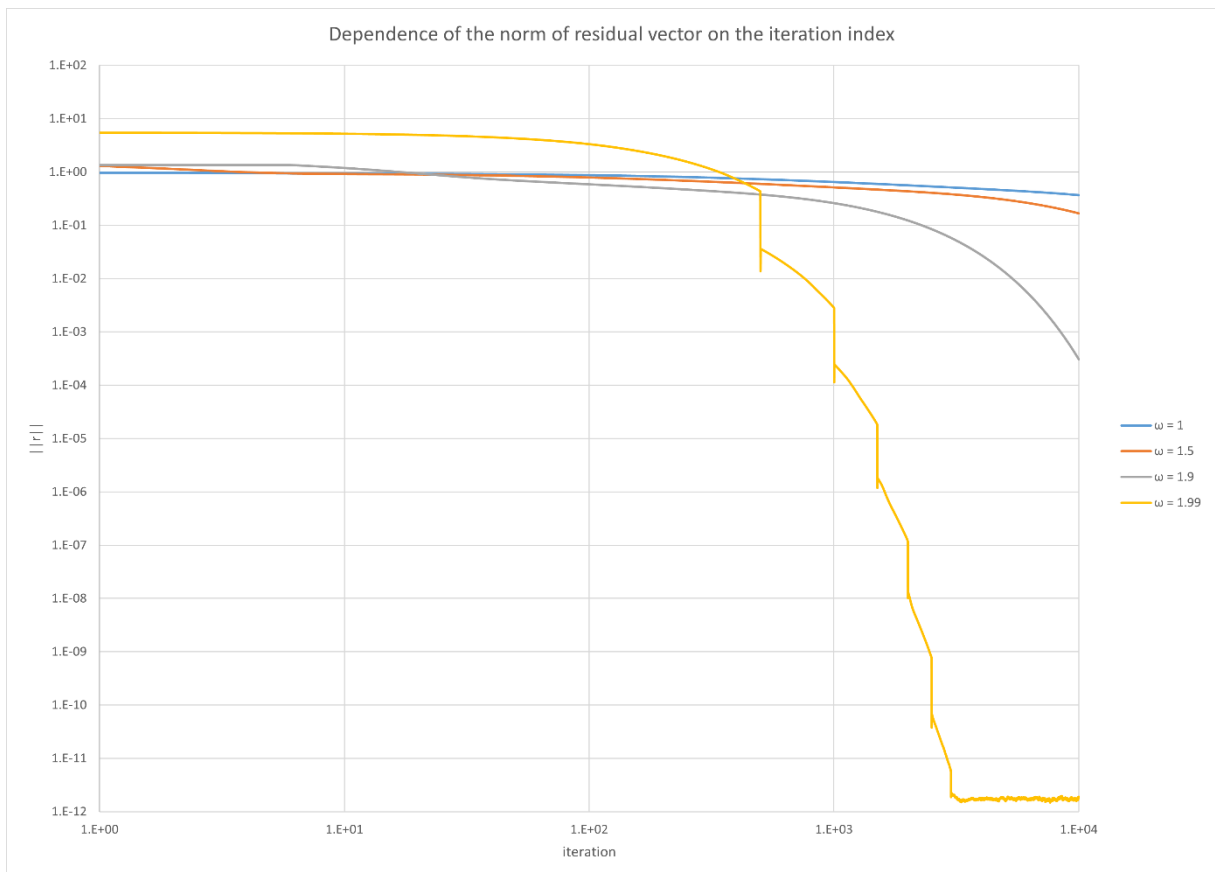
Residual vector convergence

The convergence of the residual norm for different values of the relaxation parameter is shown in Fig. 1.

For $\omega = 1$, the method converges very slowly and the residual decreases gradually. Increasing ω significantly improves convergence speed. For $\omega = 1.9$, a rapid decrease of the residual is observed.

The fastest convergence is obtained for $\omega = 1.99$, where the residual norm drops to values close to machine precision. A step-like behavior of the curve is visible, indicating strong over-relaxation.

Figure 1. Dependence of the residual norm $||\vec{r}||$ on the iteration index for different values of the relaxation parameter ω .



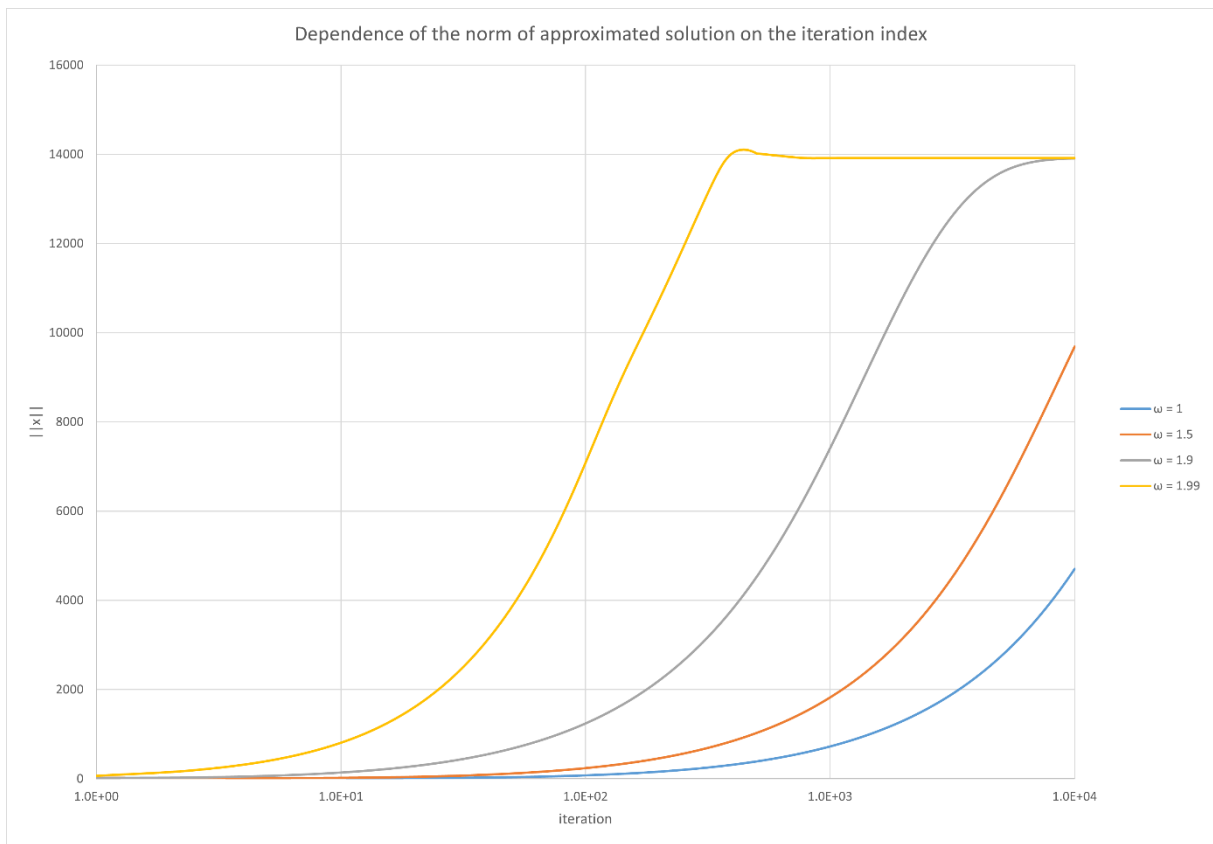
Norm of the solution

The norm of the approximated solution for different values of ω is presented in Fig. 2.

For all tested parameters, the solution converges to the same final value (~ 14000), confirming correctness of the method. However, the rate of convergence strongly depends on ω .

For larger values of ω , the solution reaches its steady state much faster. The case $\omega = 1.99$ shows the most efficient convergence.

Figure 2. Dependence of the solution norm $||\vec{v}||$ on the iteration index for different values of ω .

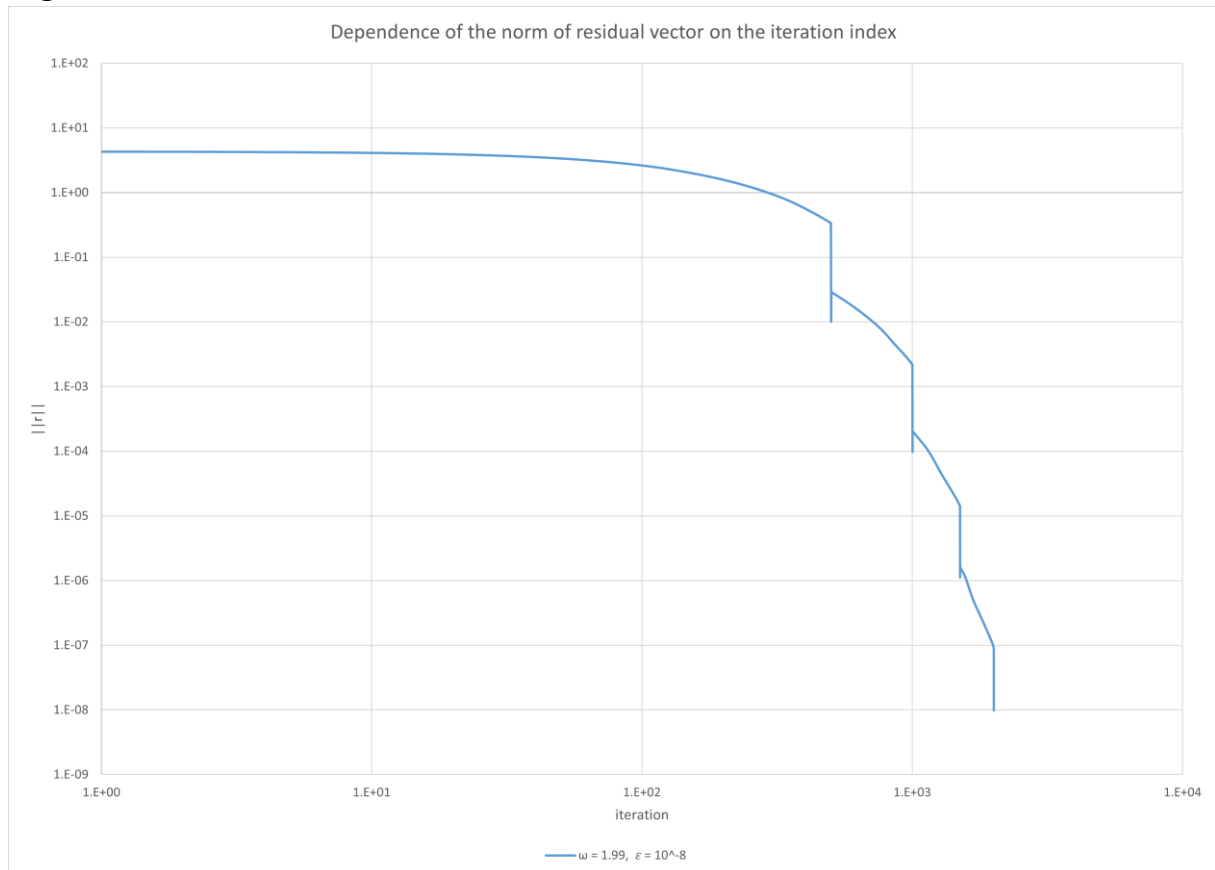


Additional test: influence of stopping criterion

An additional test was performed for $\omega = 1.99$ and a larger tolerance $\varepsilon = 10^{-8}$.

The residual norm decreases rapidly and reaches the stopping criterion after a significantly smaller number of iterations. The convergence behavior remains stable and similar in shape.

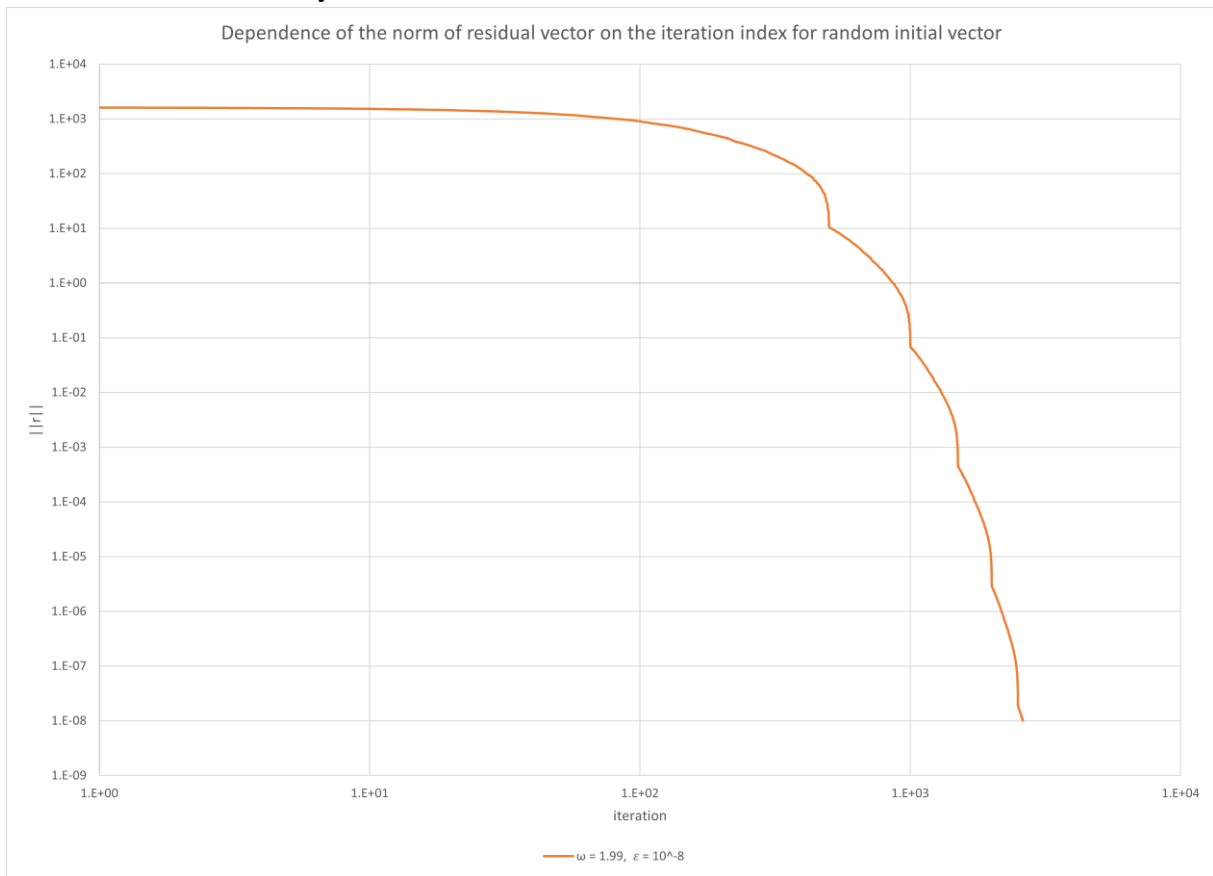
Figure 3. Residual norm calculated for $\omega = 1.99$ and $\varepsilon = 10^{-8}$.



An additional test was performed for $\omega = 1.99$ and a larger tolerance $\varepsilon = 10^{-8}$ using a randomly initialized vector.

The residual norm decreases rapidly and reaches the prescribed stopping criterion after a relatively small number of iterations. The convergence remains stable and follows quite a similar trend as in previous cases, although the initial residual is significantly larger due to a random initialization.

Figure 5. Dependence of the residual norm $\|\vec{r}\|$ on the iteration index for $\omega = 1.99$, $\varepsilon = 10^{-8}$ and randomly initialized vector $\vec{v}$.



7. Conclusions

The SOR method is an effective iterative method for solving sparse systems of linear equations arising from discretization of the Poisson equation.

The efficiency of the method strongly depends on the relaxation parameter ω . For $\omega = 1$, the convergence is very slow. Increasing ω significantly improves convergence speed, and the best performance is obtained for values closer to 2.

The case $\omega = 1.99$ provides the fastest convergence, although it introduces non-smooth (step-like) behavior in the residual decrease. All tested values of ω lead to the same final solution, which confirms correctness of the implementation.

Additional tests show that the SOR method converges for different initial approximations, including both zero and random vectors. The choice of initial vector does not affect the final solution but may influence the number of iterations required to reach the convergence.

Using a larger tolerance ε reduces the number of iterations, but also lowers the final accuracy.